AirWatch Eswatini

Environmental Monitoring System for Air Quality Prediction

USER GUIDE & SYSTEM EXPLANATION

A complete, plain-English explanation of every page of the system

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Before We Start: Understanding Air Quality

If you have never heard of air quality monitoring before, do not worry. This section explains everything from scratch, in plain language, before we look at the system itself.

What Is Air Pollution?

Air pollution means that harmful substances — things that do not belong in clean air — are floating around in the atmosphere. You cannot always see or smell these substances, but they enter your body every time you breathe. Over time, breathing polluted air causes serious health problems such as asthma, lung disease, heart attacks, and even early death.

In Eswatini, the main sources of air pollution around industrial towns are:

- Factories and manufacturing plants releasing smoke and chemicals
- Vehicles and trucks burning fuel and releasing exhaust fumes
- Burning of wood and charcoal for cooking and heating
- Burning of sugarcane fields and waste in open areas
- Agricultural activities using pesticides and chemicals

What Is PM2.5 and Why Does It Matter?

PM2.5 stands for 'Particulate Matter 2.5 micrometres'. These are tiny solid or liquid particles floating in the air. They are so small that you cannot see them individually — about 30 of them placed side by side would equal the width of a single human hair.

Why PM2.5 Is Especially Dangerous

Because PM2.5 particles are so tiny, when you breathe them in, they bypass your nose and throat — the body's natural filters — and go deep into your lungs. From there they can even enter your bloodstream. This is why PM2.5 is considered the most dangerous air pollutant by the World Health Organization (WHO).

PM2.5 is measured in micrograms per cubic metre — written as $\mu g/m^3$. This tells us how many micrograms (millionths of a gram) of these tiny particles are in each cubic metre of air.

What Is PM10?

PM10 refers to slightly larger particles — up to 10 micrometres in size. These come from dust, pollen, and smoke. They are still harmful but the body can filter some of them in the nose and throat. PM10 is tracked alongside PM2.5 to give a fuller picture of air quality.

What Are NO2 and CO?

NO2 (Nitrogen Dioxide) is a reddish-brown gas produced mainly by vehicle engines and factories. Breathing it irritates the airways and makes lung infections more likely. CO (Carbon Monoxide) is a colourless, odourless gas produced by burning fuel. It is especially dangerous because it reduces the amount of oxygen your blood can carry.

The WHO Safe Limit — The Most Important Number

The World Health Organization (WHO) has set a guideline: the annual average concentration of PM2.5 in the air should not exceed 10 µg/m³. This is considered the safe threshold above which health risks begin to appear.

> **Eswatini's Reality**
> According to WHO data and the World Bank indicator EN.ATM.PM25.MC.M3, Eswatini's annual mean PM2.5 concentration is approximately 17 µg/m³ — nearly double the WHO safe limit. In industrial towns like Matsapha, Simunye, and Bhunya, levels can be even higher on certain days.

How Do We Classify Air Quality?

The AirWatch system uses the following four categories, based on WHO PM2.5 guidelines:

Category	PM2.5 Level	What It Means	What To Do
Good	0 – 10 µg/m³	Air is clean. Safe for everyone.	No action needed. Enjoy outdoor activities.
Moderate	11 – 25 µg/m³	Air is acceptable but slightly polluted.	Sensitive people (children, elderly, asthmatic) should limit long outdoor exposure.
Unhealthy	26 – 50 µg/m³	Air quality is poor. Health effects possible	Reduce outdoor activity. Close

		for all people.	windows. Wear masks if outside.
Hazardous	Above 50 µg/m³	Very dangerous. Serious health risk for everyone.	Stay indoors. Authorities must take emergency action.

The Three Industrial Zones We Monitor

AirWatch focuses on three towns in Eswatini where industrial activity is concentrated:

Zone	Description
Matsapha	Eswatini's main manufacturing hub. Factories, industrial estates, and high vehicle traffic make this the most polluted of the three zones.
Simunye / Mhlume	An agro-processing area with sugar industry activity. Seasonal burning of sugarcane contributes to elevated pollution.
Bhunya	Home to a major paper mill. More rural than Matsapha but still affected by industrial emissions and wood burning.

SECTION 1

The Login Page

Your secure entry point into the system

What Is This Page?

The Login Page is the first thing you see when you open AirWatch Eswatini. It is a security gate — only authorised people can enter the system. Nobody can access any data, charts, or tools without first logging in with a valid username and password.

Why Do We Need a Login?

Air quality data is important information that affects public health policy, industrial regulation, and community decisions. The login system ensures that:

- Only trained and authorised staff can view and interact with the data
- Actions like uploading new data or sending alerts are tracked to a specific person
- Sensitive system settings cannot be changed by anyone who should not have access

What Are User Roles?

Not everyone who logs in sees the same things. AirWatch has three types of users, each with different levels of access:

Role	Who This Is	What They Can Do
Environmental Officer	Staff from the Eswatini Environment Authority (EEA) or environmental department	Full access: upload data, train models, run predictions, send alerts, generate reports
Researcher / Analyst	University researchers, data analysts, environmental scientists	Read-only access: view charts, historical trends, predictions, and model reports
System Administrator	The IT person responsible for managing the system	Full access plus: add/remove user accounts, manage system settings

How to Log In — Step by Step

1. Open your web browser and go to http://localhost:5173
2. You will see the AirWatch Eswatini login screen with the green leaf logo
3. Type your username in the first box (for example: officer)
4. Type your password in the second box — it will appear as dots for privacy
5. Click the eye icon on the right of the password box if you want to see what you typed
6. Click the blue 'Sign In' button
7. If your credentials are correct, you will be taken to the main dashboard

> **Dark Mode / Light Mode**
> On the login page, there is a small sun or moon icon in the top-right corner. Clicking it switches the entire application between a white (light) theme and a dark navy theme. Your preference is saved automatically so you do not have to switch every time you open the app.

SECTION 2

The Overview Page

Your dashboard — the big picture at a glance

What Is This Page?

The Overview Page is the first page you see after logging in. Think of it as your control room. It shows you the most important air quality numbers and charts all in one place, so you can quickly understand what is happening across all three industrial zones without having to dig through lots of data.

The Four Summary Cards at the Top

At the very top of the page, there are four large number cards. Here is what each one means:

Overall PM2.5 Mean

This is the average PM2.5 concentration across all three zones and all recorded dates. Think of it as the single number that summarises the overall air quality situation. Below the number, you will see how far above or below the WHO safe limit of 10 µg/m³ it is. If this number is red and shows 'above WHO', that means the air across Eswatini's industrial areas is, on average, not safe.

Most Polluted Zone

This card simply shows which of the three zones — Matsapha, Simunye, or Bhunya — has the highest average PM2.5 concentration. This helps authorities know where to focus their attention and enforcement efforts first.

Days Above WHO Limit

This percentage tells you how many of all the recorded days had PM2.5 levels above 10 µg/m³. For example, if this shows 66.9%, it means that on more than two out of every three recorded days, the air quality in at least one zone was above the safe limit. This is a powerful indicator of how chronic the pollution problem is.

Total Records

> This is simply the total number of data readings stored in the system. More records mean more reliable analysis and predictions.

The AQI Category Distribution Chart (Pie Chart)

This circular chart (called a donut chart) shows you what percentage of all recorded readings fall into each air quality category — Good, Moderate, Unhealthy, or Hazardous. Each slice of the pie is colour-coded:

- Green slice = Good air quality days (PM2.5 at or below 10 µg/m³)
- Yellow slice = Moderate days (PM2.5 between 11 and 25 µg/m³)
- Orange slice = Unhealthy days (PM2.5 between 26 and 50 µg/m³)
- Red slice = Hazardous days (PM2.5 above 50 µg/m³)

For example, if the yellow slice is the largest, it means the majority of days in the data are Moderate — above the safe limit but not dangerously so. If the red slice is large, that indicates a serious and frequent pollution problem.

The Average PM2.5 by Zone Chart (Bar Chart)

This chart shows three bars, one for each zone. The height of each bar represents the average PM2.5 concentration for that zone. A red dashed horizontal line across the chart marks the WHO safe limit of 10 µg/m³. Any bar that rises above this red line means that zone's average pollution level is unsafe.

Matsapha's bar is typically the tallest (most polluted), followed by Simunye, then Bhunya. This directly reflects the industrial intensity of each area.

The Monthly PM2.5 Heatmap

This is a grid with the three zones listed down the left side, and the twelve months of the year across the top. Each cell in the grid is colour-coded from green (clean) through yellow and orange to red (heavily polluted), and shows the average PM2.5 for that zone during that month.

What to look for: you will notice that cells for May, June, July, August, and September tend to be more orange or red. This is because Eswatini's dry season runs from May to September, during

which biomass burning (wood fires, sugarcane burning) increases significantly, raising pollution levels. The wet season months (October to April) tend to show greener cells because rain washes particles out of the air.

The WHO Warning Banner

At the bottom of the Overview page there is an amber warning banner reminding users that Eswatini's national PM2.5 mean of approximately 17 µg/m³ exceeds the WHO guideline. This is not an alarm — it is a persistent reminder of the baseline situation, sourced from the World Bank and WHO databases.

SECTION 3

Historical Trends Page

Exploring pollution patterns over time

What Is This Page?

The Historical Trends page lets you explore how air pollution has changed over time. Instead of just seeing average numbers, here you can see the actual ups and downs of pollution levels day by day across different time periods. It helps answer questions like: Is pollution getting worse over the years? Which months are always bad? Is one zone consistently more polluted than the others?

The Zone Filter Buttons

At the top of the page, there are three coloured buttons — one for each zone (Matsapha, Simunye, Bhunya). By clicking these buttons, you can choose which zones to show on the chart. You can show all three at once, or isolate just one zone to study it on its own. When a zone is selected, the button fills with its colour (blue for Matsapha, purple for Simunye, green for Bhunya).

The Pollutant Selector

Next to the zone buttons, there is a dropdown menu where you can switch between different pollutants to display on the chart. The options are:

- PM2.5 — The main pollutant of concern (tiny particles, most dangerous)
- PM10 — Slightly larger particles (dust, pollen, smoke)
- NO2 — Nitrogen dioxide from vehicles and factories
- CO — Carbon monoxide from burning fuel

The Time Series Line Chart

The main chart on this page is a line graph showing how the selected pollutant has changed over time. Each zone has its own coloured line. The horizontal axis (bottom) shows dates. The vertical axis (left side) shows the concentration of the pollutant in $\mu g/m^3$.

When PM2.5 is selected, a red dashed horizontal line appears at 10 µg/m³ — the WHO safe limit. Any part of a zone's line that is above this red dashed line represents a period when that zone's air quality was unsafe.

> **How to Read the Line Chart**
> If Matsapha's blue line spikes sharply upward during July and August, that tells you those months regularly have the worst air quality in Matsapha — likely due to dry season burning. If the line stays flat and low, air quality is stable and relatively clean. Sudden sharp spikes can indicate industrial incidents or unusual burning events.

Dry vs Wet Season Comparison (Bar Chart)

Below the time series chart is a grouped bar chart that compares average PM2.5 levels during the dry season (May to September) versus the wet season (October to April) for each zone. Orange bars represent the dry season, blue bars represent the wet season.

This chart almost always shows the same pattern: the dry season bars are significantly taller than the wet season bars. This is because during the dry season there is no rain to clear the air, and more biomass is burned for heating. Understanding this seasonal pattern helps authorities plan when to issue public warnings.

Zone PM2.5 Summary (Progress Bars)

On the right side of the page are horizontal progress bars, one for each zone. These give you a quick visual comparison of the average PM2.5 level per zone. The longer the bar, the higher the average pollution. Below each bar, the minimum and maximum values are shown, giving you a sense of how much the readings vary — a big gap between min and max means pollution levels swing a lot; a small gap means levels are fairly consistent.

SECTION 4

Upload & Validate Dataset Page

Bringing real air quality data into the system

What Is This Page?

This page is only available to Environmental Officers and System Administrators. It is where new air quality data is brought into the system. Whenever the EEA or another environmental organisation collects new readings from monitoring stations, those readings need to be imported here so the system can analyse and predict from them.

What Format Does the Data Need to Be In?

The data must be in a CSV file. CSV stands for 'Comma-Separated Values' — it is a simple spreadsheet format that can be created or exported from Microsoft Excel, Google Sheets, or any data collection software. The file must have these columns in the header row:

Column Name	Example Value	What It Means
`date`	`2024-01-15`	The date the reading was taken. Must be in YYYY-MM-DD format.
`location`	`Matsapha`	Which industrial zone. Must be exactly: Matsapha, Simunye, or Bhunya.
`pm25`	`18.5`	PM2.5 concentration in µg/m³. A decimal number.
`pm10`	`32.0`	PM10 concentration in µg/m³. Optional but recommended.
`no2`	`15.0`	Nitrogen dioxide in µg/m³. Optional but recommended.
`co`	`0.5`	Carbon monoxide in mg/m³. Optional but recommended.

The Validation and Audit Process

Before the system accepts the data, it runs an automatic validation check. This is like a quality control step that catches problems before they affect the analysis. Here is what the system checks:

- Required columns check: Does the file have all the necessary column names?
- Location check: Are all location values exactly 'Matsapha', 'Simunye', or 'Bhunya'? Any spelling mistake will be flagged.
- Value range check: Are PM2.5 values within the realistic range of 0 to 500 µg/m³?
- Null/missing values check: How many readings are missing? If more than 30% of a column is empty, a warning is shown.
- Duplicate check: Are there rows with the same date and location? Duplicates will be removed.

If there are critical errors (like missing columns or wrong location names), the system will not accept the file and will show a red error message explaining exactly what needs to be fixed. If there are only warnings (like some missing values), the system will accept the file but tell you what it automatically fixed.

What Happens After Validation?

Once the file passes validation, the officer clicks 'Process and Save Dataset'. The system then cleans the data (fills in missing values, removes duplicates, adds the AQI category to each row) and saves it to the database. This processed data then becomes the foundation for all the charts, trends, and predictions in the rest of the system.

SECTION 5

Configure & Train Models Page

Teaching the computer to predict air quality

What Is This Page?

This is where the artificial intelligence (AI) that powers the predictions is trained. Do not be intimidated by the word 'training' — in this context it simply means: we feed the computer a large amount of historical air quality data, and it learns the patterns in that data so that it can predict future values.

Think of it like teaching someone to predict tomorrow's weather by studying years of weather records. After seeing thousands of examples, they start to notice patterns — 'when the wind blows from the east in July, it usually stays dry' — and can make educated guesses about what will happen next. The computer does the same thing, but with air quality data.

The Two Machine Learning Models

AirWatch uses two different mathematical approaches to predict PM2.5 levels. Using two models and comparing them is standard practice in data science — it helps ensure the predictions are reliable.

Model 1: Random Forest

A Random Forest is like asking hundreds of different experts the same question and taking the most common answer. Technically, it builds 200 separate decision trees (like flowcharts), each one trained on a slightly different version of the data. Each tree gives a prediction, and the forest's final answer is the average of all 200 trees. This approach is very robust — one tree making a mistake does not spoil the whole result.

Model 2: SVR (Support Vector Regression)

SVR is a different mathematical technique that tries to draw the best possible line through the data points, allowing for a small margin of error. It is particularly good at handling complex, non-linear relationships — situations where increasing one variable does not simply and directly increase another. SVR is more sensitive to the scale of the data, which is why the system pre-scales all the numbers before training it.

The Configurable Parameters — What They Mean

Environmental Officers can adjust several settings before training. Here is what each one means in plain language:

Random Forest Settings

- Number of Trees (50 to 500): How many decision trees to build. More trees = more accurate but slower to train. 200 is a good balance.
- Max Depth (3 to 20): How deep each decision tree can grow. Deeper trees can spot more complex patterns but may become too specific to the training data (this is called overfitting). 12 is a safe default.
- Min Leaf Samples (1 to 20): The minimum number of data points a tree branch needs before it can split further. Higher values prevent the tree from making decisions based on just one or two unusual readings.

SVR Settings

- C (Regularisation): Controls how strictly the model fits the data. Higher C = tighter fit but more risk of overfitting. Lower C = more flexible but may miss patterns. 100 is the recommended default.
- Epsilon: The size of the 'tolerance tube' around the prediction line. Predictions within this tube are considered acceptable. 0.5 µg/m³ means the model accepts being half a microgram off.
- Kernel: The mathematical shape used to separate data points. 'RBF' (Radial Basis Function) works well for most non-linear data and is the recommended choice.

What Features Does the Model Learn From?

The models are not just given raw PM2.5 readings. The preprocessing step creates additional features that give the models more context:

- PM2.5 from yesterday (lag1): The single most powerful predictor — if yesterday was polluted, today probably is too
- PM2.5 from 3 days ago (lag3): Short-term trend information
- PM2.5 from 7 days ago (lag7): Week-level pattern — weekly cycles in industrial activity
- 7-day rolling average (roll7): Smoothed recent trend
- 30-day rolling average (roll30): Longer-term trend context

- Month, year, day of week: Calendar-based patterns
- Is dry season: Whether the reading falls in the high-pollution dry season
- Zone indicators: Which zone the reading belongs to

What Happens After Training?

After training, the system automatically evaluates both models on data they have never seen before (the test set — 20% of the dataset reserved for testing). It then calculates three performance metrics:

- MAE (Mean Absolute Error): The average difference in µg/m³ between the model's predictions and the actual recorded values. Lower is better. An MAE of 3.29 µg/m³ means the model is typically within 3.29 µg/m³ of the true value.
- RMSE (Root Mean Squared Error): Similar to MAE but penalises large errors more heavily. It shows how badly the model performs on its worst predictions.
- R^2 (R-Squared): A score from 0 to 1 (or expressed as a percentage) showing how much of the variation in PM2.5 the model successfully explains. An R^2 of 0.68 means the model explains 68% of why PM2.5 goes up and down.

The model with the higher R^2 is automatically saved as the 'best model' and used for all future predictions. Both models are saved so they can be compared in the Model Report page.

SECTION 6

Predict Air Quality Page

Forecasting tomorrow's air quality today

What Is This Page?

The Predict Air Quality page is the heart of the entire system. It is where the trained machine learning model is put to work. An officer enters today's measured pollutant readings, and the system predicts what PM2.5 levels will look like over the next 3 to 14 days for a chosen industrial zone.

Think of it like a weather forecast, but for air pollution. Just as a weather app tells you it will rain on Thursday, AirWatch tells you that Matsapha's PM2.5 is predicted to reach Unhealthy levels on Thursday — giving authorities time to act before it happens.

The Input Form — What You Need to Enter

Before getting a forecast, the officer must enter the current measurements. Here is what each field means:

- Industrial Zone: Select which zone you want to forecast — Matsapha, Simunye, or Bhunya
- Forecast Days: How many days ahead to predict (3, 5, 7, 10, or 14 days)
- Current PM2.5: Today's measured PM2.5 concentration in µg/m³
- Current PM10: Today's measured PM10 concentration in µg/m³
- Current NO2: Today's measured NO2 concentration in µg/m³
- Current CO: Today's measured CO concentration in mg/m³

Where Do These Numbers Come From?

In a fully operational monitoring system, these readings would come from sensors deployed in each industrial zone. In the current implementation, the officer enters the most recent readings from whatever monitoring equipment is available, including readings from low-cost sensors deployed by Afri-Set in Matsapha and readings downloaded from international air quality databases.

How Does the Forecast Work?

When you click 'Run Forecast', the following happens behind the scenes:

8. The system takes your current readings and the current date
9. It builds the full set of features the model needs: lag values, rolling averages, season flags, zone flags
10. It feeds these features into the trained Random Forest (or SVR) model
11. The model returns a predicted PM2.5 value for Day 1
12. That Day 1 prediction is then used as an input for Day 2's prediction, and so on
13. This chain continues for the number of days you selected

Reading the Results — The 7-Day Cards

After running the forecast, a row of cards appears — one for each forecasted day. Each card shows:

- The abbreviated day name (Mon, Tue, Wed, etc.)
- The date (month and day)
- The predicted PM2.5 value in large, bold text
- The AQI category in the matching colour (green for Good, yellow for Moderate, orange for Unhealthy, red for Hazardous)

The colour-coding makes it immediately obvious which days are expected to be safe and which are dangerous — you do not need to read any numbers to understand the overall picture.

The Forecast Line Chart

Below the cards is a line chart showing the same data in graph form. The horizontal axis shows the dates, the vertical axis shows PM2.5 levels, and the red dashed horizontal line marks the WHO safe limit of 10 µg/m³. Any predicted point above the red line represents a day when air quality is expected to be unsafe.

The Warning Alert

If any day in the forecast is predicted to reach Unhealthy levels (above 25 µg/m³), an orange warning banner automatically appears at the bottom of the page. It identifies the peak day and its predicted value, and prompts the officer to go to the Email Notifications page to alert

stakeholders. If all days are expected to stay below 25 µg/m³, a green success message is shown instead.

SECTION 7

Generate Report Page

Creating professional summaries for decision-makers

What Is This Page?

The Generate Report page automatically assembles a professional environmental report using all the data currently in the system. This report is designed to be shared with government officials, environmental regulators, industrial managers, and community representatives who need a clear summary of air quality conditions — without having to navigate through charts themselves.

What Does the Report Contain?

The report is divided into three sections:

1. Zone Summary Table

A table showing, for each of the three industrial zones:

- Total number of records analysed
- Average PM2.5 concentration
- Maximum recorded PM2.5 (worst single day)
- Average PM10 concentration
- Percentage of days where PM2.5 exceeded the WHO limit
- A status label — either 'Exceeds WHO' (shown in red) or 'Within WHO' (shown in green)

2. Recommendations

Based on the data, the system automatically generates a set of actionable recommendations:

- Identifies the most polluted zone and recommends prioritising emission control efforts there
- States the overall percentage of days above the WHO limit
- Recommends proactive early warnings during the dry season (May to September)
- Recommends expanding continuous monitoring infrastructure
- Recommends community awareness programmes for affected residents

3. Export Options

The report can be exported as a CSV spreadsheet for further analysis in Excel, or printed directly from the browser as a PDF by pressing Ctrl+P and selecting 'Save as PDF'. This makes it easy to share the report by email or present it in a meeting.

SECTION 8

Email Notifications Page

Alerting stakeholders when air quality is dangerous

What Is This Page?

When the Predict Air Quality page forecasts dangerous PM2.5 levels, somebody needs to be told. The Email Notifications page allows the Environmental Officer to send a professional, formatted email alert directly to stakeholders — government officials, community leaders, health departments, or industrial managers — directly from within the system.

How Does It Work?

The page automatically pulls in the most recent forecast generated on the Predict page. It displays the 7-day forecast cards so the officer can review what is being sent. The officer then:

14. Enters their Gmail address (this is the address the email will be sent from)
15. Enters their Gmail App Password (a special password created for third-party apps — not the same as the normal Gmail password)
16. Enters the recipient email addresses, one per line (for example, the EEA director, the health department, and the affected factory manager)
17. Optionally customises the email subject line
18. Clicks 'Send Notification'

> **What Is a Gmail App Password?**
>
> A Gmail App Password is a 16-character code generated in your Google Account settings. It allows applications like AirWatch to send emails through Gmail without storing your actual Gmail password. To create one: go to myaccount.google.com → Security → 2-Step Verification → App Passwords. This is a standard, secure approach used by many professional applications.

What Does the Email Look Like?

The recipient receives a neatly formatted HTML email that includes:

- The AirWatch Eswatini header with the system name
- The zone and date the forecast was generated

- A table showing each forecasted day, its PM2.5 value, and its AQI category
- A reminder that the WHO safe limit is 10 µg/m³
- A footer identifying the email as coming from the EMCU environmental monitoring system

What If Email Sending Is Not Available?

If the officer does not have a Gmail App Password set up, or if there is no internet connection, there is a fallback option at the bottom of the page: a button to download the forecast as a CSV file, which can then be attached to a manually composed email or printed and distributed in person.

SECTION 9

Model Report Page

Understanding how well the AI is performing

What Is This Page?

The Model Report page gives a detailed, technical (but clearly explained) view of how well the two machine learning models — Random Forest and SVR — are performing. It is designed for researchers, analysts, and supervisors who want to understand the reliability of the predictions.

The Metric Cards

For each model, three metrics are displayed:

R^2 Score (R-Squared)

This is the most important metric. It tells you what percentage of the variation in PM2.5 levels the model successfully explains. An R^2 of 0.68 means the model correctly accounts for 68% of why PM2.5 goes up and down. The remaining 32% is due to factors the model cannot capture (like unexpected industrial accidents or unusual weather events). A higher R^2 means a more reliable model.

MAE (Mean Absolute Error)

This is the average difference in µg/m³ between what the model predicted and what actually happened. An MAE of 3.29 µg/m³ means that on average, the model's prediction is within 3.29 µg/m³ of the true value. For context: the WHO safe limit is 10 µg/m³, so being off by 3.29 is significant but acceptable for a forecasting tool operating on limited data.

RMSE (Root Mean Squared Error)

RMSE is similar to MAE but it punishes large errors more severely. If the model occasionally makes a prediction that is very far off, RMSE will be much higher than MAE. A RMSE close to the MAE indicates the model makes consistently similar-sized errors (which is good). A much higher RMSE than MAE means the model sometimes makes very large mistakes.

The Comparison Bar Chart

A grouped bar chart compares all three metrics side by side for both models. Blue bars represent Random Forest, purple bars represent SVR. The best model is highlighted with a green 'Best' badge. In the AirWatch system, Random Forest typically outperforms SVR on this dataset because the lag features (especially yesterday's PM2.5) create a strong linear relationship that RF captures very efficiently.

The Interpretation Section

At the bottom of the page, a written interpretation summarises the key findings:

- Which model performed best and its exact R^2 score
- What the MAE means in practical terms
- That the single most powerful predictor is yesterday's PM2.5 reading (lag1), which accounts for approximately 79% of the Random Forest's decision-making
- That seasonal patterns and rolling averages contribute additional meaningful accuracy
- That both models consistently confirm Eswatini's industrial zones exceed the WHO PM2.5 threshold

SECTION 10

Manage Users Page

Controlling who has access to the system

What Is This Page?

The Manage Users page is only visible to the System Administrator. It is the control panel for managing who can log into AirWatch and what they are allowed to do. From here, the administrator can see all current users, add new ones, and remove old accounts.

The Current Users List

The page displays a table of all registered users showing their username, full name, and role. Each user also has a delete button (the red bin icon) next to their name. The administrator cannot delete their own account — this prevents accidentally locking everyone out of the system.

Adding a New User

To give a new person access to the system, the administrator fills in a form with:

- Username: The short name the person will type when logging in (example: jdoe)
- Full Name: Their real name as it will appear in reports (example: John Doe)
- Password: A temporary password — the person should change this after their first login
- Role: Select whether they are an Environmental Officer, Researcher/Analyst, or System Administrator

Once submitted, the new user's account is immediately active and they can log in with the provided credentials.

Removing a User

If a staff member leaves or no longer needs access, the administrator selects their name from a dropdown and clicks 'Remove User'. This permanently deletes their account and they will no

longer be able to log in. All data they uploaded or predictions they ran remain in the system — removing a user only removes their login credentials.

SECTION 11

How the System Works — The Technology

A plain-English explanation of the architecture

The Three-Layer Structure

AirWatch Eswatini is built in three separate layers that work together. Understanding how they connect helps explain why the system works the way it does.

Layer	Technology Used	Plain English Role
Frontend (User Interface)	React + Vite + Tailwind CSS	The visual part you see and click on. Built with React, a modern JavaScript library used by companies like Facebook and Airbnb.
Backend API	Python Flask	The middle layer that handles requests, runs the ML models, and communicates between the interface and the database.
Database	MySQL via XAMPP	Where all the air quality data, user accounts, and prediction logs are permanently stored.
Machine Learning	scikit-learn (Python)	The library that provides the Random Forest and SVR algorithms. Industry standard, used by data scientists worldwide.

The Data Journey — From Reading to Prediction

Here is the full journey that air quality data takes through the system:

19. A monitoring sensor records PM2.5, PM10, NO2, and CO readings at a specific location and time
20. An Environmental Officer downloads or collects these readings as a CSV file
21. The officer uploads the CSV through the Upload & Validate page
22. The system validates the data, cleans it, and stores it in the MySQL database
23. The Configure & Train page is used to train the Random Forest and SVR models on this historical data
24. The Predict page takes today's readings, applies the trained model, and forecasts the next 7 days
25. If dangerous levels are predicted, the Email Notifications page sends alerts to stakeholders

26. The Generate Report page compiles all findings into a shareable document
27. Administrators monitor everything through the Overview and Historical Trends pages

SECTION 12

Data Source & Limitations

Being honest about what the system can and cannot do

Where Does the Training Data Come From?

A key challenge in developing this system is that continuous, publicly available air quality monitoring data for Eswatini's specific industrial zones does not yet exist in structured digital form. The Eswatini Environment Authority (EEA) has limited continuous monitoring infrastructure, and national air quality data is sparse and not regularly published.

To overcome this challenge while maintaining academic integrity, the system was developed and tested using a real-world air quality dataset from India (sourced from Kaggle — a public data science platform), which contains several years of daily PM2.5, PM10, NO2, and CO readings from multiple industrial cities. Three Indian cities with pollution profiles similar to Eswatini's industrial zones were selected:

- Ahmedabad (mapped to Matsapha) — heavy manufacturing, highest pollution
- Lucknow (mapped to Simunye) — mid-level agro-industrial activity
- Shillong (mapped to Bhunya) — smaller, more rural industrial profile

The pollution values from these cities were then scaled down using calibration factors derived from WHO and World Bank data for Eswatini (specifically the World Bank indicator EN.ATM.PM25.MC.M3), so that the resulting dataset reflects Eswatini's published national PM2.5 mean of approximately 17 µg/m³. The seasonal patterns, diurnal trends, and pollutant relationships in the data are genuinely real — only the geographic labels and scale have been adapted.

Academic Disclosure Statement

This approach — using proxy data from comparable regions when local data is unavailable — is a well-established and academically accepted methodology in environmental science research, particularly in developing countries where monitoring infrastructure is limited. The proposal itself acknowledges data scarcity as a core limitation of current environmental monitoring in Eswatini.

Limitations of the System

Every system has limitations. Being transparent about them is important for responsible use:

- Prediction accuracy: The Random Forest model achieves an R^2 of approximately 68%, meaning it cannot explain all the variation in PM2.5. Real-world events like unexpected factory emissions or unusual weather events can cause deviations from predictions.
- Data quality: The system is only as good as the data fed into it. If the uploaded readings contain errors or are from malfunctioning sensors, predictions will be unreliable.
- No real-time sensor integration: In the current implementation, data must be manually uploaded by an officer. Full automation would require direct sensor API connections.
- Geographic limitation: The system currently only covers Matsapha, Simunye, and Bhunya. Expanding to other areas would require new training data.
- Internet dependency: Email notifications require an active internet connection and a configured Gmail App Password.

Recommendations for Future Development

- Deploy low-cost air quality sensors directly in each industrial zone and connect them to the system via API for real-time data ingestion
- Partner with the Eswatini Environment Authority to access any historical monitoring data they hold
- Expand the geographic coverage to include urban areas like Mbabane and Manzini
- Add weather data integration (wind speed, humidity, rainfall) as additional prediction features
- Develop a mobile app version for officers in the field
- Incorporate transboundary pollution data from Mpumalanga, South Africa, which significantly influences Eswatini's air quality

AirWatch Eswatini

Environmental Monitoring System for Air Quality Prediction in Industrial Areas

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